

Comparative Risk Aversion (The Arrow-Pratt Theorem)

For $t = 0, 1$, let u_t be a C^2 vN-M utility with $u'_t > 0$ on $\mathbb{R}_{++}$. It follows that there is a C^2 function $T : \text{Range}(u_0) \rightarrow \mathbb{R}$ such that $u_1(z) = T(u_0(z))$ for all $z > 0$ and $T' > 0$.

Let $\mathcal{L}$ be the set of cumulative distribution functions on $\mathbb{R}$ with $F(0) = 0$ and $F(z) = 1$ for some number $z < \infty$.¹ For $t = 0, 1$, denote the Arrow-Pratt absolute risk aversion measure by $r_t(z) = -\frac{u''_t(z)}{u'_t(z)}$, the certainty equivalent by $CE_t(F)$ and the risk premium by $RP_t(F)$.

Theorem. *The following are equivalent.*

1. $r_1(z) \geq r_0(z)$ for all $z > 0$.
2. The function T is **concave**.
3. $CE_1(F) \leq CE_0(F)$ for all $F \in \mathcal{L}$.
4. $RP_1(F) \geq RP_0(F)$ for all $F \in \mathcal{L}$.

Corollary. *Let u, F satisfy the hypotheses of the Theorem, and let $CE(F, a)$ be the certainty equivalent and $RP(F, a)$ the risk premium for vN-M utility u when the positive number a is added to the risk z (so final wealth is $z + a$): $u(CE(F, a)) = \int u(z + a)dF(z)$ and $RP(F, a) = \int (z + a)dF(z) - CE(F, a)$. The following are equivalent*

1. $r(\cdot)$ is decreasing on $\mathbb{R}_{++}$.
2. $CE(F, a) - a \geq CE(F, 0)$ for all $F \in \mathcal{L}$ and $a > 0$.
3. $RP(F, a) \leq RP(F, 0)$ for all $F \in \mathcal{L}$ and $a > 0$.

To prove the corollary, set $u = u_1$, define $u_0(z) = u(z + a)$, for $a > 0$ and apply the Theorem. An important boundary case is *constant* (absolute) risk aversion: $u(z) = -e^{-\lambda z}$ for $\lambda > 0$.

The corollary can be extended to *proportional* or *relative* lotteries. Set $z = wy$, where the distribution F of y is in $\mathcal{L}$ and $w > 0$. Define $v(y; w) = u(wy)$. Now apply the Theorem and to $v(\cdot; w)$ with parameter w . An increase in w leads to less aversion over lotteries over y if $-v_{yy}/v_y$ is decreasing in w . Use the definition of v to find that this last property is equivalent to $-zu''(z)/u'(z)$ decreasing, which is called *decreasing relative risk aversion*. An important case used in applications is *constant relative risk aversion*: $-zu''(z)/u'(z) = 1 - \gamma > 0$, which is equivalent (up to an increasing affine transformation) to $u(z) = \gamma^{-1}z^\gamma$ for $1 > \gamma \neq 0$.

¹In words, $\mathcal{L}$ is the set of cdfs on $\mathbb{R}_+$ with bounded support. The last condition just ensures that the integral $\int u(z)dF(z)$ is finite and that “the derivative of the integral is the integral of the derivative” whenever u is C^1 .